Classical Field Theory — Exercise Sheet 15

Beyond the one-dimensional cubic proof

Hints and answers

Use the mass-one Klein–Gordon group $U_d(t)$ on $\mathcal{E}_d = H^1(\mathbb{R}^d) \times L^2(\mathbb{R}^d)$, with Hilbert norm $\|(f,g)\|_{\mathrm{lin}}^2 = \|f\|_{H^1}^2 + \|g\|_2^2$. Lecture 15 proves the forced estimate in one dimension; Problem 2 checks its dimension-independent Fourier proof. All fields below are real.

Exercise 15.1 (A general potential and an affine background). Let $V \in C^3(\mathbb{R})$ and let b be a smooth bounded solution of $-b'' + V'(b) = 0$ on $\mathbb{R}$. Do not assume $b \in L^2$. For $\phi = b + v$ consider $\phi_{tt} - \phi_{xx} + V'(\phi) = 0$.

(a) Derive the mass-one perturbation force $N_b(v)$, and show that it maps $H^1(\mathbb{R})$ locally Lipschitzly to $L^2(\mathbb{R})$, vanishing at zero. Give a constant on a ball of radius R using the maximum of $|V''|$ on a specified compact interval.

(b) For every initial ball $\|(v_0, g_0)\|_{\mathrm{lin}} \le B$, $B \ge 1$, give a common contraction radius and positive time. Prove existence, uniqueness among weak energy solutions, continuous dependence, and the finite-lifespan alternative. Which derivatives of V did the estimates use?

Hint. Subtract the stationary equation before asking for an L^2 force. On the bounded field range use the integral of V'' along the segment between two field values, then use Lecture 15's forcing lemma.

Answer.

$$N_b(v) = v - [V'(b+v) - V'(b)], \qquad L_R = 1 + \max_{|s| \le \|b\|_\infty + R/\sqrt{2}} |V''(s)|.$$

Take radius $2B$ and time $1/(2L_{2B})$. The estimate needs only $V \in C^2$.

Exercise 15.2 (The cubic survives in three dimensions). Take as given the Sobolev estimate $\|u\|_{L^6(\mathbb{R}^3)} \le C_S \|u\|_{H^1(\mathbb{R}^3)}$ for every real $u \in H^1$, where C_S is a fixed positive constant. Consider

$$u_{tt} - \Delta u + u^3 = 0, \qquad (u(0), u_t(0)) \in \mathcal{E}_3.$$

(a) Justify the three-dimensional version of the forced mass-one energy estimate, including the weak equation. Prove local existence, energy-class uniqueness and continuous dependence by a Duhamel contraction, with a common lifespan for a bounded initial ball.

(b) Would the same $H^1 \to L^2$ force argument handle u^5? For a nonzero real $\eta \in C_c^\infty(\mathbb{R}^3)$, compute the scaling of $u_\Lambda(x) = \Lambda^{1/2}\eta(\Lambda x)$, $\Lambda \ge 1$, in H^1, and of its cubic and quintic forces in L^2. Limit the conclusion to this contraction argument.

Hint. In Fourier space the linear rotation uses $\nu(k) = \sqrt{1 + |k|^2}$ in every dimension. For the cubic difference apply Hölder with three L^6 factors rather than an L^∞ factor.

Answer. The reference force is $N(u) = u - u^3$, with $L_R = 1 + 3C_S^3 R^2$. For $B \ge 1$, take $R = 2B$, $T = 1/(2L_R)$. The quintic L^2 norm scales as Λ, while the cubic norm is fixed and the H^1 norms stay bounded.

Exercise 15.3 (Sine-Gordon on a weighted path space). Let b be a smooth bounded stationary sine-Gordon background, $-b'' + \sin b = 0$, and write $\phi = b + v$. For any fixed $T > 0$, equip $X_T = C([0,T]; \mathcal{E}_1)$ with the equivalent norm

$$\|z\|_a = \sup_{0 \le t \le T} e^{-at} \|z(t)\|_{\mathrm{lin}}, \qquad a > 0.$$

Derive the reference mass-one Duhamel force. Show that the Duhamel map is a contraction on the entire space X_T for $a = 4$, with no radius restriction. Prove existence and uniqueness on every finite interval and give a continuous-dependence bound. Why is an amplitude-dependent local constant unnecessary? Do not assume that b or $\sin b$ belongs to L^2.

Hint. The derivative of $v - \sin(b+v) + \sin b$ lies between zero and two. The weighted norm estimates a time integral by integrating $e^{-a(t-s)}$.

Answer. $N_b(v) = v - \sin(b+v) + \sin b$ has global Lipschitz constant two into L^2. The weighted contraction factor is at most $2/a = 1/2$. For fixed data differences the unweighted solution difference is at most $2e^{4T}$ times their energy norm.

Exercise 15.4 (Defocusing continuation with the analytic input stated). For the three-dimensional cubic solution of Problem 2, assume additionally that the following energy identity holds throughout its maximal energy-class lifespan:

$$E_0 = \int_{\mathbb{R}^3} \left[\frac{1}{2} u_t^2 + \frac{1}{2} |\nabla u|^2 + \frac{1}{4} u^4 \right] \mathrm{d}^3 x.$$

This is an additional hypothesis at the weak energy regularity, not a consequence of L3's classical balance. Prove that the initial integral is finite and that the solution is global. Give a bound for its full $\mathcal{E}_3$ norm on every finite time interval, explicitly recovering the L^2 displacement term.

Hint. Interpolate L^4 between L^2 and the supplied L^6 bound. The missing L^2 estimate follows from the norm primitive of u_t.

Answer.

$$\|(u(t), u_t(t))\|_{\mathrm{lin}}^2 \le \left(\|u(0)\|_2 + |t| \sqrt{2E_0} \right)^2 + 4E_0.$$

The continuation alternative therefore excludes every finite endpoint.